

Frida ke Zariye SSL Pinning Bypass – Mobile Se Network Hacking

Frida ek dynamic instrumentation toolkit hai jo mobile applications mein runtime par code inject karne ki facility deta hai. Android apps built-in SSL/TLS certificate pinning check ko bypass karne ke liye isko use karte hain. Is method ke zariye app ki certificate verification ko hijack kar ke, app ko apne intercepting certificate (jaise Burp/Charles) ko trust karne par majboor kiya jata hai aur HTTPS traffic capture ho jata hai.

Requirements

- Rooted Android device ya emulator
- USB Debugging enabled
- ADB (Android Debug Bridge) installed
- Python 3.x with pip
- Burp Suite ya koi bhi intercepting proxy

Step 1: Frida Setup

Sab se pehle computer par Frida install karein. Python package install karne ke liye `pip install frida-tools` command use karein. Installation ke baad `frida --version` se version check kar sakti hain.

Iske baad device ki CPU architecture check karein. `adb shell getprop ro.product.cpu.abi` command se architecture pata chalega. Architecture ke hisaab se frida releases se server download karein. arm64-v8a ke liye `frida-server-android-arm64`, armeabi-v7a ke liye `frida-server-android-arm`, x86_64 ke liye `frida-server-android-x86_64`, aur x86 ke liye `frida-server-android-x86` download karein. Computer aur phone par Frida versions match hone chahiye, warna errors aa sakte hain.

Ab device par Frida Server deploy karein. `adb push frida-server /data/local/tmp/` se server push karein, `adb shell "chmod 755 /data/local/tmp/frida-server"` se permissions set karein, aur `adb shell "/data/local/tmp/frida-server &"` se server start karein. Connection verify karne ke liye `frida-ps -U` use karein.

Step 2: Proxy Setup

Burp Suite mein proxy listener configure karein. Proxy tab → Proxy Settings → Proxy Listeners mein jaayein aur add listener with port 8080 karein. Bind to address mein "All interfaces" select karein.

Mobile device par proxy set karein. Settings → Wi-Fi → modify network mein jaayein. Proxy ko Manual par set karein. Proxy hostname mein PC ki IP address daalein aur Proxy port 8080 daalein.

Burp certificate install karein. Mobile browser mein <http://burpsuite> visit karein aur certificate download karein. Settings → Security → Install from storage mein install karein. Android 7+ mein user-installed certificates apps trust nahi karti, isliye system certificate store mein install karne ke liye root access required hai.

Step 3: Frida Script Inject Karein

Pehle frida-ssl-bypass.js script save karein. Yeh script Java perform function mein SSL bypass logic implement karti hai. Script Burp certificate ko load karti hai, KeyStore create karti hai, TrustManager banati hai, aur SSLContext init method ko hijack karti hai.

Script mein CertificateFactory, FileInputStream, X509Certificate, KeyStore, TrustManagerFactory, aur SSLContext classes use hoti hain. Burp certificate ko /data/local/tmp/cert-der.crt path se load kiya jata hai aur custom TrustManager banaya jata hai jo humare certificate ko trust kare.

Ab Burp certificate ko DER format mein convert karein aur device par push karein. openssl x509 -in certificate.cer -outform DER -out cert-der.crt se convert karein aur adb push cert-der.crt /data/local/tmp/ se push karein. Phir adb shell "chmod 644 /data/local/tmp/cert-der.crt" se permissions set karein.

Script inject karne ke liye spawn mode use karein agar app start ke saath inject karna hai toh: frida -U -f com.example.app -l frida-ssl-bypass.js --no-pause. Agar app already running hai toh attach mode use karein: frida -U -n "com.example.app" -l frida-ssl-bypass.js. App package name find karne ke liye adb shell pm list packages | grep keyword use karein.

Common SSL Pinning Bypass Techniques

Script different SSL pinning implementations ko bypass kar sakti hai. Android API mein `X509TrustManager.checkClientTrusted` override aur `TrustManagerImpl.verifyChain` hook karti hai. Square OkHTTP library mein `CertificatePinner.check` implementation override karti hai. `WebViewClient` mein `WebViewClient.onReceivedSslError` handler override karti hai. Android 7+ Network Security Config policy override karti hai. Native level (Flutter, OpenSSL, BoringSSL) ke liye `libflutter.so` mein native function hooking required hai.

Common Issues Aur Solutions

Frida Connection Error aata hai toh USB debugging enabled check karein, ADB connection verify karein `adb devices` se, aur versions match karein.

Proxy not working ho toh Burp listener "All interfaces" par set hai ensure karein, mobile aur PC same network mein hain check karein, aur Intercept ON hai confirm karein.

Certificate not trusted ho toh Android 7+ par user certificate install karna kaam nahi karta. System certificate store mein install karein ya Magisk `TrustUserCerts` module use karein.

App detects Frida toh FridaBypassKit script ya stealth-ssl-bypass-anti-detection.js use karein.

Flutter app bypass ke liye Flutter BoringSSL use karta hai, isliye native hooking required hai.

Frida Commands Reference

Frida Server start karne ke liye command hai `adb shell "/data/local/tmp/frida-server &"`.

Spawn mode mein script inject karne ke liye `frida -U -f com.example.app -l script.js --no-pause` use karein.

Attach mode mein script inject karne ke liye `frida -U -n "com.example.app" -l script.js` use karein.

Running processes list karne ke liye `frida-ps -U` use karein.

Device par installed packages list karne ke liye `adb shell pm list packages | grep keyword` use karein.

Certificate device par push karne ke liye `adb push cert-der.crt /data/local/tmp/` use karein.

 **Important Security Note:** Yeh process sirf educational aur authorized penetration testing ke liye hai. Kisi doosre ke network ya device par bina permission ke traffic intercept karna applicable cyber laws ke under punishable hai. Rooting device se warranty void ho sakti hai aur security model compromised ho jata hai.

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